

Employee Request Automation System

Solution Summary | Mohammed Saif Shirgaonkar

Objective. Replace fragmented employee request handling with a standardized and trackable workflow that automatically classifies, routes, stores, monitors, and escalates requests across HR, IT, Payroll, Operations, and General Support.

1. Solution Overview

The Proof of Concept provides a centralized employee request workflow. Employees submit requests through a standardized web portal. FastAPI manages validation and business logic, OpenAI classifies each free-text request, MongoDB Atlas stores the ticket, n8n performs team routing and scheduled SLA automation, and Slack delivers operational notifications.

Each request receives a unique ticket ID and follows a controlled lifecycle: Open -> Active -> Finalized. The prototype is hosted on Render so external reviewers can test the system without relying on the developer's local environment.

2. Foundational Assumptions

- The standardized web portal is the implemented intake channel for the PoC.
- Requests are classified into HR, IT, Payroll, Operations, or Other.
- Priority is Critical, High, Medium, or Low, with target SLAs of 1, 4, 8, and 24 hours respectively.
- Each category maps to a responsible support team, and Slack is the operational notification channel.
- Finalized requests require no further SLA monitoring.
- An unresolved overdue ticket should receive continued attention without being re-notified every 15 minutes.
- Email, SMS, WhatsApp, and other messaging channels are future intake extensions rather than implemented PoC integrations.

3. Architectural Justification

Component	Design rationale
FastAPI	Central application layer for validation, business rules, ticket lifecycle, web routes, and secured internal automation APIs.
OpenAI	Used only where interpretation is required: converting free-text requests into category and priority. Team assignment and SLA remain deterministic application rules.
MongoDB Atlas	Persistent ticket storage for request details, classification output, status, SLA deadlines, and escalation timestamps.
n8n	Separates operational automation from application logic through new-request routing and scheduled SLA monitoring/escalation.
Slack	Provides immediate team visibility without requiring continuous dashboard monitoring.
Render	Hosts the assessment PoC publicly so reviewers can access it independently of the local development environment.

4. Where AI & Automation Deliver the Highest ROI

The highest-value use of AI is at the request triage stage. Employees can describe an issue naturally instead of needing to know the correct department or urgency level. Automated classification creates a consistent starting point for routing while reducing manual triage.

Automation	Operational value
Automatic routing	New tickets are immediately sent to the responsible team.
SLA calculation	Target deadlines are determined automatically from priority.
Team notification	n8n sends each new ticket to the corresponding Slack channel.
Overdue detection	A scheduled workflow checks every 15 minutes for unresolved tickets beyond SLA.
Controlled escalation	Overdue tickets are routed back to the responsible team's Slack channel.
Spam prevention	The escalated_at timestamp limits escalation to one notification per ticket per calendar day, while allowing another reminder the next day if unresolved.

5. Operational Control & Resolution

The Request Management Dashboard gives centralized visibility into ticket classification, priority, assigned team, SLA deadline, and current status. Support users can progress requests through Open -> Active -> Finalized. Finalized tickets are excluded from SLA monitoring.

The SLA workflow calls dedicated FastAPI internal endpoints protected by an X-Internal-API-Key. Application credentials are stored as environment variables rather than committed to source control. This keeps the PoC simple while separating public request submission from internal automation operations.

6. Scalability Roadmap

- Omni-channel intake: normalize email, SMS, WhatsApp, or enterprise messaging requests into the same processing layer.
- Identity and access: add authentication/SSO, role-based access control, and team-specific work queues.
- Operational governance: introduce full ticket audit history, assignment history, and controlled status transitions.
- Analytics: report response time, resolution time, SLA breach rate, request volume, and team performance.
- Resilience: use asynchronous queues and retries for higher volume and failure isolation.
- Observability: add centralized logging, metrics, alerting, and workflow monitoring.
- Production delivery: introduce managed secrets, CI/CD, automated integration tests, and highly available infrastructure.

Design principle. AI is focused on the ambiguous part of the process - interpreting employee language. Routing, SLA management, lifecycle transitions, and escalation controls remain deterministic and auditable.